

# Project Plan

PINN\_M3DC1

Physics-informed surrogate architecture for M3D-C1-shaped nonlinear resistive-MHD workflows

**Purpose.** This is the living roadmap for PINN\_M3DC1. The project begins with a verified two-dimensional reduced-MHD benchmark. One surrogate is then trained first with simulation data alone and then with the reduced-MHD equations and known conditions added to its loss. Genuine M3D-C1 exports enter only through a later adapter and require a new validation claim.

**Current status.** Notes for Modules 1, 2, and 3 are complete. Module 2 combines simulation, checking, saving, and data grouping. Module 3 combines the former data-only and physics-informed surrogate modules because the baseline is simply the same model with the physics-loss weights set to zero. Validation and M3D-C1 transfer are now Modules 4 and 5. No MHD solver, data pipeline, or trained neural network is yet implemented.

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## 1 Objective and claim boundary

The objective is to determine whether physics-informed learning improves a parameterized neural surrogate for nonlinear resistive-MHD evolution under controlled data budgets. Version 1 uses independently generated synthetic arrays. It supports the claim that a PINN can approximate a stated reduced model; it does not establish fidelity to M3D-C1. A validated M3D-C1 surrogate requires genuine code outputs, complete metadata, held-out M3D-C1 cases, and renewed verification.

## 2 Initial physics benchmark

The periodic two-field model evolves magnetic flux  $\psi$  and stream function  $U$ . Define

$$[f, g] = f_x g_y - f_y g_x, \quad J = -\nabla^2 \psi, \quad \omega = \nabla^2 U. \quad (2.1)$$

The frozen version-1 residuals are

$$R_\psi = \partial_t \psi + [U, \psi] - \eta \nabla^2 \psi, \quad (2.2)$$

$$R_\omega = \partial_t \omega + [U, \omega] - [J, \psi] - \nu \nabla^2 \omega. \quad (2.3)$$

The first nonlinear case is magnetic-island coalescence. It is small enough for independent implementation while retaining advection, current-sheet formation, resistive reconnection, dissipation, and parameter-dependent time evolution.

## 3 Module roadmap

| No. | Module                                        | Purpose                                                                                                                                                                           |
|-----|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Reduced Resistive-MHD Physics and Conventions | Freeze variables, signs, normalization, boundary conditions, invariants, and diagnostic definitions.                                                                              |
| 2   | Generate and Save MHD Simulation Data         | Run checked island-coalescence cases, save the numerical arrays and settings, and divide complete simulations into training, validation, and test groups.                         |
| 3   | Physics-Informed Surrogate                    | Use one model for both settings: $\mathcal{L}_{\text{data}}$ alone is the baseline; adding the two PDE residuals, the initial condition, and the gauge makes it physics-informed. |
| 4   | Validation, Diagnostics, and Benchmarking     | Compare field accuracy, physical histories, residuals, generalization, runtime, and individual loss-term tests.                                                                   |
| 5   | M3D-C1 Export Adapter and Transfer Learning   | Map genuine exported fields and metadata into the project contract without silently changing physical meaning.                                                                    |

## 4 Data and model contract

A synthetic case stores  $(x, y, t)$ ,  $\psi$ ,  $U$ ,  $J$ ,  $\omega$ ,  $\eta$ ,  $\nu$ ,  $A_0$ , the initial conditions, and the solver settings. The neural-network task is

$$(\psi_0, U_0, \eta, \nu, t) \mapsto (\hat{\psi}(t), \hat{U}(t)), \quad (4.1)$$

with  $\hat{J} = -\nabla^2 \hat{\psi}$  and  $\hat{\omega} = \nabla^2 \hat{U}$ . Plot images are not training data because they do not contain the complete numerical field values.

## 5 Controlled comparison

| Model                              | Data loss | Physics losses        | Role                     |
|------------------------------------|-----------|-----------------------|--------------------------|
| Reduced-MHD solver                 | –         | Discretized equations | Synthetic reference      |
| Module 3, data-only setting        | yes       | no                    | Fair neural baseline     |
| Module 3, physics-informed setting | yes       | PDE/IC/gauge          | Physics-information test |

The two neural models use identical case splits, observations, and comparable capacity. Entire  $(\eta, \nu, A_0)$  combinations are excluded from training. Interpolation and extrapolation are reported separately.

## 6 Verification and validation

The reference solver must pass derivative, bracket, periodicity, divergence, convergence, and energy-behavior checks before producing training data. Surrogate evaluation includes relative  $L_2$  and  $L_\infty$  errors for all fields, peak-current and current-sheet errors, magnetic and kinetic energy histories, reconnection metrics, PDE residuals, and periodicity. Runtime reporting includes reference generation, training, inference, and break-even query count.

## 7 Implementation sequence

1. Freeze the equations and signs.
2. Implement the solver in Python and MATLAB.
3. Produce one checked island-coalescence simulation and save it.
4. Generate more cases and divide complete simulations into training, validation, and test groups.
5. Train the Module 3 model using  $\mathcal{L}_{\text{data}}$  alone.
6. Turn on the two PDE residual losses, the initial-condition loss, and the gauge loss.
7. Test optional constraints one at a time, then run generalization and cost analysis.
8. Add and validate an adapter only when genuine M3D-C1 arrays are available.

## 8 Acceptance milestones

**M1 – synthetic solver.** One command writes a reproducible HDF5 case and diagnostics; refinement and dissipation checks pass.

**M2 – fair learning experiment.** The data-only and physics-informed settings of Module 3 share cases, samples, model assumptions, and scoring.

**M3 – held-out validation.** Field and physics metrics are reported on unseen parameter combinations, with interpolation separated from extrapolation.

**M4 – M3D-C1 transfer.** Exported variables, mesh, normalization, closures, and boundary conditions are mapped explicitly; the full test suite is rerun before an M3D-C1 surrogate claim is made.